

Probability and Statistics – for Computer Science

Based on Forsyth, *Probability and Statistics for Computer Science* (2018)

Applied Statistics and Probability for Computer Scientists

Experiments

Chapter 8 — One-Way ANOVA, Two-Way ANOVA, and Experimental Design

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Chapter Outline | What we will cover

Why ANOVA?

One-Way ANOVA

Post-Hoc Tests & Assumptions

Two-Way ANOVA

Interactions & Effect Sizes

Experimental Design Principles

Summary

Learning Objectives

- ▶ Explain why comparing many groups needs **ANOVA**, not repeated t -tests
- ▶ Partition variance and compute the **F -statistic** for one- and two-way designs
- ▶ Interpret **interactions** between two factors
- ▶ Apply the **three R's** of experimental design

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Section 1

Why ANOVA?

The Problem with Multiple t -Tests | Comparing 5 groups needs 10 pairwise tests

The multiple testing problem

With k groups, pairwise t -tests require $\binom{k}{2}$ comparisons — each at risk of a false positive.

The solution: ANOVA

Analysis of Variance tests *all* groups at once with a single F -test: "are any of the means different?"

Section 2

One-Way ANOVA

The One-Way ANOVA Model | One factor, k groups

Hypotheses

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

H_1 : at least one μ_i differs

Example

Compare crop yield under $k = 4$ fertilizers — one factor (fertilizer type), four levels.

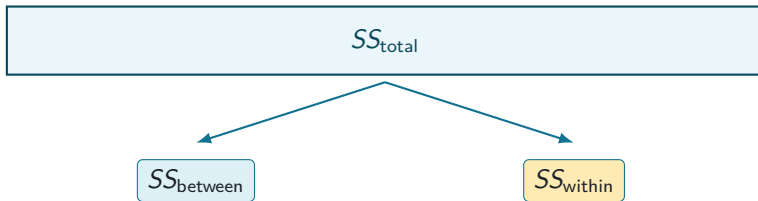
Partitioning Variance | The core idea behind every ANOVA

The key relationship

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}$$

Between vs. within

- ▶ SS_{between} : groups differ from the grand mean
- ▶ SS_{within} : individuals differ from their group mean



The F -Statistic | Signal (between) vs. noise (within)

Mean squares

$$MS_B = \frac{SS_B}{k-1}, \quad MS_W = \frac{SS_W}{N-k}$$

Test statistic

$$F = \frac{MS_B}{MS_W} \sim F_{k-1, N-k}$$

Logic

If H_0 is true, between- and within-group variation should be similar ($F \approx 1$). A large F means groups differ more than chance alone would predict.

The ANOVA Table | The standard summary format

Source	SS	df	MS	F	p
Between groups	SS_B	$k - 1$	$SS_B / (k - 1)$	MS_B / MS_W	–
Within groups	SS_W	$N - k$	$SS_W / (N - k)$		
Total	SS_T	$N - 1$			

Section 3

Post-Hoc Tests & Assumptions

Post-Hoc Tests: Which Groups Differ? | A significant F doesn't say *which* pair

The problem

ANOVA's F -test only tells you *some* group differs — not which one(s).

Tukey's HSD

Honestly Significant Difference — compares all pairs while controlling the overall (family-wise) error rate.

Three key assumptions

- ▶ Independent observations
- ▶ Approximately normal residuals within each group
- ▶ Equal variance across groups (**homogeneity**)

Eta-squared η^2

$$\eta^2 = \frac{SS_B}{SS_T}$$

The fraction of total variance explained by group membership — the ANOVA analogue of R^2 .

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Section 4

Two-Way ANOVA

The model

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij}$$

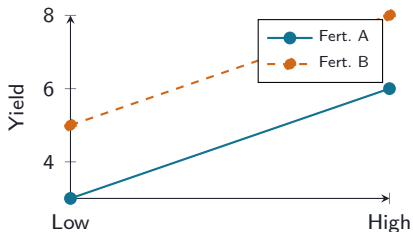
Three hypotheses tested at once

- ▶ Main effect of factor A
- ▶ Main effect of factor B
- ▶ **Interaction** between A and B

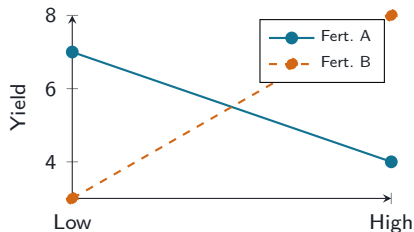
Why not run two separate one-way ANOVAs?

A single two-way design is more efficient *and* it's the only way to detect an interaction between the factors.

Understanding Interactions | Do the factors' effects depend on each other?



Watering level
Parallel lines: **no interaction**



Watering level
Crossing lines: **interaction present**

Section 5

Interactions & Effect Sizes

Interpreting an Interaction | Read the interaction before the main effects

Rule of thumb

If the interaction is significant, interpret main effects *cautiously* — the effect of one factor depends on the level of the other.

Simple effects analysis

Follow up by examining the effect of one factor *separately at each level* of the other.

Balanced design

Equal sample size in every cell of the factor combination — simplifies the sum-of-squares decomposition and is easier to interpret.

Partial eta-squared

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{error}}}$$

Effect size for one term, controlling for the others.

Section 6

Experimental Design Principles

The Three R's of Experimental Design | The foundation of trustworthy experiments

Principle	Purpose
Randomization	Spread unknown confounds evenly across groups
Replication	Estimate and shrink the effect of random noise
Blocking	Remove the effect of a known nuisance variable

Without randomization

Any systematic difference between groups (e.g. assigning healthier patients to the new drug) will bias the result.

Blocking: The Randomized Complete Block Design | Controlling a known source of variation

The idea

Group experimental units into **blocks** that are similar (e.g. by field, by day), then randomize treatments *within* each block.

Why it helps

Removes block-to-block variation from the error term, making real treatment effects easier to detect.

Common Experimental Designs | Matching design to structure

Design	When to use
Completely randomized (CRD)	No known nuisance factor
Randomized complete block (RCBD)	One known nuisance factor
Latin square	Two nuisance factors, limited runs
Factorial	Studying two or more factors (and interactions) together



Section 7

Summary

Before you experiment

- ▶ Define hypotheses and effect size of interest in advance
- ▶ Randomize treatment assignment
- ▶ Plan enough replication for adequate power

Common mistakes

Confounding a treatment with time or location; too few replicates; ignoring a known blocking variable.

Compare many groups

ANOVA partitions variance into between- and within-group components and tests with a single F -statistic.

Two factors

Two-way ANOVA adds **interactions** — the effect of one factor can depend on another.

Design matters

Randomization, replication, and blocking make the *analysis* trustworthy in the first place.

Looking ahead

Ch. 9: **inferring** the parameters of probability models themselves — MLE and Bayesian inference.

Thank you!

Questions & Discussion

Next: Chapter 9 — Inferring Probability Models